

CASTEAU, Belgium

WMO: 064325

Lat: 50.503N

Lon: 3.983E

Elev: 78

StdP: 100.39

Time Zone: 1.00 (EUC)

Period: 10-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.9	-3.8	-10.1	1.6	-5.7	-7.0	2.1	-1.6	8.4	9.6	7.5	9.3	2.2	40	0.468

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.2	30.0	19.9	27.9	18.9	25.9	17.9	20.7	27.7	19.7	26.1	18.7	23.8	2.6	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	13.3	22.5	17.8	12.9	21.9	16.9	12.2	20.9	60.1	28.9	56.5	26.2	53.5	23.8	24.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
6.8	5.9	5.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
			DB												
			WB	-8.7	22.9	3.0	0.9	-10.8	23.5	-12.5	24.1	-14.2	24.6	-16.3	25.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.2	3.8	4.0	7.3	10.5	13.3	17.1	19.0	18.3	15.2	12.1	7.6	5.1	(6)
(7)		DBStd	6.38	3.79	4.05	3.43	3.89	3.68	3.24	3.16	2.99	2.86	3.55	3.70	3.85	(7)
(8)		HDD10.0	768	193	168	95	39	8	0	0	0	0	19	90	157	(8)
(9)		HDD18.3	2749	451	401	342	235	161	63	30	36	103	194	322	412	(9)
(10)		CDD10.0	1189	1	1	12	55	111	212	279	257	156	84	18	4	(10)
(11)		CDD18.3	128	0	0	0	1	6	25	51	34	9	1	0	0	(11)
(12)		CDH23.3	1111	0	0	0	17	59	236	428	289	75	6	0	0	(12)
(13)		CDH26.7	311	0	0	0	0	11	63	123	96	18	0	0	0	(13)
(14)	Wind	WSAvg	2.3	2.7	2.7	2.5	2.2	2.1	2.0	2.0	1.9	2.1	2.3	2.7	(14)	
(15)	Precipitation	PrecAvg	809	70	62	53	48	60	76	76	85	61	68	76	78	(15)
(16)		PrecMax	1011	162	138	137	120	117	156	149	190	171	131	126	167	(16)
(17)		PrecMin	629	4	16	3	0	11	28	22	16	14	11	9	21	(17)
(18)		PrecStd	98	36	28	31	28	27	30	35	51	35	28	28	35	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	15.8	19.9	25.8	28.8	31.9	33.0	33.1	29.1	24.0	18.1	14.1	(19)
(20)			MCWB	10.5	9.1	11.4	15.6	17.9	21.1	20.6	20.7	18.7	16.9	14.3	11.3	(20)
(21)		2%	DB	11.8	12.0	17.8	23.0	25.1	28.2	29.9	29.1	25.2	21.1	16.0	13.0	(21)
(22)			MCWB	10.0	8.8	11.1	13.6	16.3	19.3	19.5	19.8	17.6	16.3	13.6	11.2	(22)
(23)		5%	DB	10.1	10.2	15.0	20.1	22.9	26.0	27.8	26.2	22.2	18.9	14.2	11.9	(23)
(24)			MCWB	8.7	8.4	9.8	13.1	15.3	18.1	18.7	18.5	16.9	14.8	12.0	10.0	(24)
(25)		10%	DB	9.0	9.2	12.8	17.0	20.2	23.9	25.8	24.0	20.1	17.0	13.0	10.2	(25)
(26)			MCWB	7.9	7.8	9.1	11.7	14.5	17.4	18.0	17.5	15.2	14.3	11.2	8.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.5	10.4	12.6	16.0	19.7	21.7	21.6	22.5	20.1	17.8	15.7	12.4	(27)
(28)			MCDWB	12.4	12.1	18.0	23.6	26.0	28.2	28.7	29.7	27.1	22.5	17.5	13.7	(28)
(29)		2%	WB	10.0	9.6	11.2	14.2	17.8	20.2	20.5	20.4	18.7	16.4	13.8	11.3	(29)
(30)			MCDWB	10.9	10.9	16.0	20.8	22.1	26.6	27.8	27.3	23.9	20.3	15.7	12.8	(30)
(31)		5%	WB	8.9	8.8	10.4	13.1	16.2	18.8	19.6	19.3	17.5	15.4	12.5	10.3	(31)
(32)			MCDWB	9.9	10.3	14.0	18.9	21.3	24.2	25.9	24.9	21.5	18.2	14.3	11.7	(32)
(33)		10%	WB	7.7	7.8	9.6	12.1	15.0	17.8	18.6	18.3	16.2	14.6	11.3	9.1	(33)
(34)			MCDWB	8.7	9.4	12.4	17.2	20.1	22.4	23.8	22.8	19.6	16.9	13.0	10.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.5	5.4	7.7	8.9	9.1	9.4	9.2	9.1	8.4	6.8	5.2	4.5	(35)
(36)			MCDBR	5.0	6.9	12.4	12.9	12.5	13.2	12.7	12.9	11.7	9.1	6.1	4.6	(36)
(37)			MCWBR	4.4	5.0	6.7	6.1	5.8	5.8	5.0	5.6	5.8	5.0	4.5	3.5	(37)
(38)		5% WB	MCDBR	5.1	6.2	10.2	11.6	10.7	11.5	11.0	11.2	10.2	8.1	5.2	4.4	(38)
(39)			MCWBR	4.9	4.9	6.2	5.9	5.5	5.5	4.9	5.6	5.4	4.7	4.2	3.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.326	0.347	0.382	0.416	0.427	0.424	0.425	0.416	0.388	0.364	0.338	0.323	(40)	
(41)		taud	2.395	2.357	2.258	2.180	2.194	2.226	2.225	2.272	2.337	2.402	2.395	2.366	(41)	
(42)		Ebn at Noon	701	779	811	823	831	836	827	815	798	749	675	639	(42)	
(43)		Edh at Noon	64	84	111	133	137	134	132	121	102	80	63	57	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.81	1.49	2.65	4.20	4.94	5.41	5.14	4.37	3.32	1.96	0.95	0.64	(44)	
(45)		RadStd	0.10	0.29	0.36	0.47	0.47	0.50	0.57	0.44	0.30	0.17	0.11	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (71 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page